

ECE 60146 HW10

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1 Training Losses

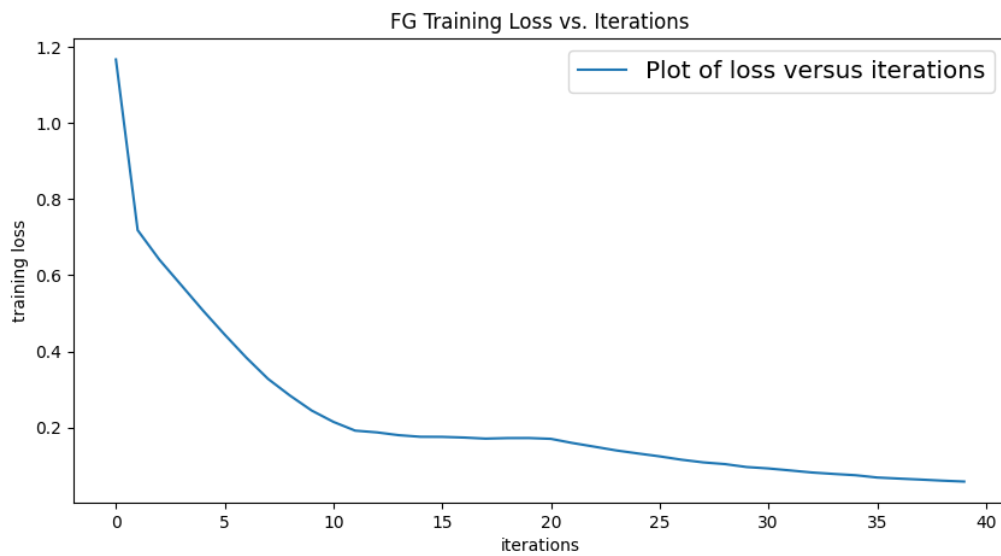


Figure 1: Training Loss for transformerFG

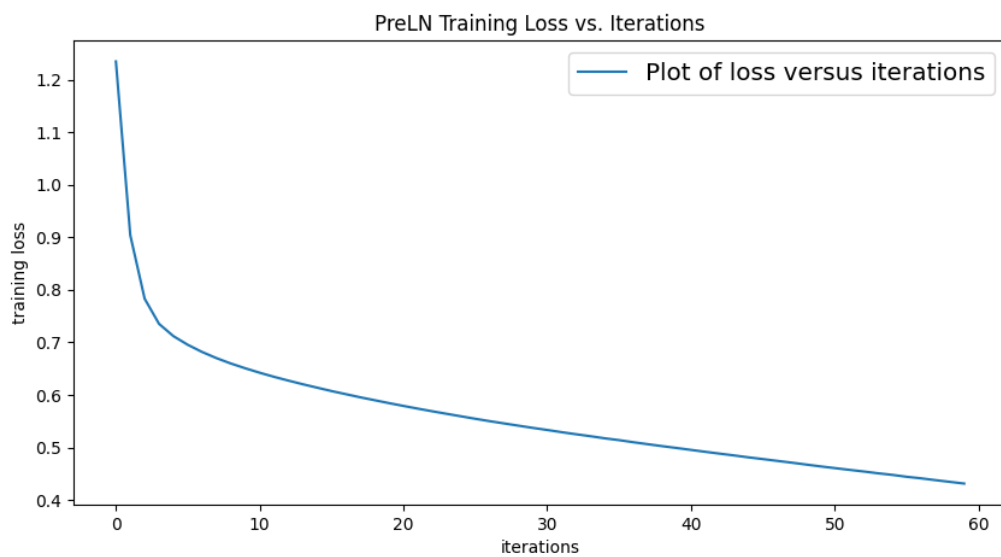


Figure 2: Training Loss for trasformerPreLN

2 Output Examples

Input English Phrase	Translated Ground Truth	Translated trans-formerFG	Translated trans-formerPreLN
SOS we are pretty much in agreement EOS	SOS estamos bastante de acuerdo EOS	EOS estamos bastante de acuerdo EOS EOS EOS EOS EOS	SOS estamos EOS EOS EOS EOS EOS EOS EOS
SOS i want to live in a big city EOS	SOS quiero vivir en una gran ciudad EOS	EOS quiero vivir en una gran EOS EOS EOS EOS EOS	SOS quiero EOS EOS EOS EOS EOS EOS EOS
SOS this pen does not write well EOS	SOS esta pluma no escribe bien EOS	EOS esta pluma no escribe bien EOS EOS EOS EOS EOS	SOS esta EOS EOS EOS EOS EOS EOS EOS EOS
SOS we are all in the same boat EOS	SOS todos estamos en el mismo barco EOS	EOS todos estamos en mismo mismo barco EOS EOS EOS	SOS todos EOS EOS EOS EOS EOS EOS EOS EOS
SOS i have never played tennis with tom EOS	SOS nunca he jugado al tenis con tom EOS	EOS nunca he jugado tenis tenis con tom EOS EOS EOS	SOS nunca EOS EOS EOS EOS EOS EOS EOS EOS

Table 1: Five Examples of Outputs from Both Models

3 Levenshtein Distance

3.1 Code

```

1 import numpy as np
2
3 def report_statistics(distances):
4     print(f"Mean: {np.mean(distances):.2f}")
5     print(f"Median: {np.median(distances):.2f}")
6     print(f"Standard Deviation: {np.std(distances):.2f}")
7     print(f"Maximum: {np.max(distances):.2f}")
8     print(f"Minimum: {np.min(distances):.2f}")
9
10 def clean_sentence(sentence):
11     return [word for word in sentence if word not in ("SOS", "EOS", "EOS\n", "SOS\n")]
12
13 def extract_results(file):
14     with open(file, "r", encoding="utf-8") as file:
15         lines = file.readlines()
16
17     gt = []
18     pred = []
19     for i in range(0, len(lines), 4):
20         input_line = lines[i+2]
21         start1 = input_line.find("[") + 1
22         end1 = input_line.find("]") + 1
23         start2 = input_line.find("[", start1) + 2
24         end2 = input_line.find("]", end1) - 1
25         gt.append(clean_sentence(input_line[start2:end2].split('\n')))
26
27         output_line = lines[i+3]
28         output_str = output_line.split(':')[1]
29         pred.append(clean_sentence(output_str[2:].split('\n')))
30     return gt, pred
31
32 def levenshtein_distance(str1, str2):
33     # Ensure str1 is the longer string
34     if len(str1) < len(str2):
35         str1, str2 = str2, str1
36
37     len_str1, len_str2 = len(str1), len(str2)

```

```

38
39 # Initialize two rows for dynamic programming
40 previous_row = list(range(len_str2 + 1))
41 current_row = [0] * (len_str2 + 1)
42
43 for i in range(1, len_str1 + 1):
44     current_row[0] = i
45     for j in range(1, len_str2 + 1):
46         cost = 0 if str1[i - 1] == str2[j - 1] else 1
47         current_row[j] = min(
48             previous_row[j] + 1,      # Deletion
49             current_row[j - 1] + 1,    # Insertion
50             previous_row[j - 1] + cost # Substitution
51         )
52     previous_row, current_row = current_row, previous_row
53
54     return previous_row[-1]
55
56 def get_statistics(file):
57     gt, pred = extract_results(file)
58     dist = []
59     for i in range(len(gt)):
60         dist.append(levenshtein_distance(gt[i], pred[i]))
61     report_statistics(dist)
62
63 print("Report Statistics from FC model")
64 get_statistics("translations_with_FG_40.txt")
65
66 print("\nReport Statistics from PreLN model")
67 get_statistics("translations_with_PreLN_60.txt")

```

Model	Mean	Median	Std. Dev.	Max	Min
FC Model	0.83	1.00	1.01	5.00	0.00
PreLN Model	4.58	5.00	1.61	8.00	1.00

Table 2: Levenshtein distance statistics for FC and PreLN models

The Levenshtein Distance results show that the FC model performed significantly better than the PreLN model. The FC model had a lower average distance (0.83 vs. 4.58), and its median distance was also lower (1.00 compared to 5.00). These statistics make sense because from the example outputs, we can see that the FG model got most of the words correct. However, the FG model does have extra trailing "EOS" words. On the other hand, the PreLN model did not perform well. It was only able to output one correct word, with the rest of the words being "EOS".

The standard deviation for the FC model was smaller (1.01 vs. 1.61), indicating that its performance was more consistent. It also managed to produce exact matches in some cases, with a minimum distance of 0.00, while the PreLN model's lowest distance was 1.00. On the other hand, the PreLN model had a higher maximum distance of 8.00, showing that some of its translations were quite far off.